UNIT 3 · TOPIC 8

Selection and Management of Farm Machinery

Syllabus topic: Farm machinery selection and management for different soils, crops and operations

Selection factors

Type	Factors
Technical	Soil type (heavy clay → higher draft, disc vs MB choice; wetland → cage wheels/puddlers); crop and operation sequence; farm size and field shape; available power source (human/animal/power tiller/tractor); implement–tractor compatibility (hitch category, PTO power, hydraulic capacity)
Managerial/economic	TIMELINESS (finish within the optimum window); annual use expected; custom-hiring availability; dealer/spares service; operator skill; subsidy/credit

$$\text{Capacity needed} = \text{area} \div (\text{working days} \times \text{h/day} \times \text{probability of good weather})$$

Management functions

- Scheduling of operations.
- Maintenance planning (daily/periodic service schedules).
- Record keeping (hours, fuel, repairs) — the basis of cost control and replacement decisions.
- Operator training.
- Machine **replacement** when rising repair costs + obsolescence exceed the value of the remaining service (economic life).
- **Custom Hiring Centres (CHCs)** for high-cost machines in small-holding agriculture (India).

Timeliness sizing, step by step

Example. 40 ha must be sown within 10 days, 8 h/day; drill field efficiency 75% at 4 km/h.
1. Required EFC = 40/(10 × 8) = **0.5 ha/h**.
2. TFC = 0.5/0.75 = **0.667 ha/h**.
3. Width = 10 × TFC/S = 10 × 0.667/4 ≈ **1.7 m** → choose a 9-row × 20 cm (**1.8 m**) drill.

Practice MCQs

Q122. Machine replacement is economically indicated when:

(a) Paint fades
(b) Rising repair cost + obsolescence exceed value of remaining service
(c) After 1 year
(d) Never

Q123. Timeliness sizing: 40 ha in 10 days × 8 h/day needs an effective capacity of:

(a) 0.25
(b) 0.5
(c) 1.0
(d) 2.0 ha/h

Q124. For EFC 0.5 ha/h at η_f 75%, required TFC is about:

(a) 0.375
(b) 0.5
(c) 0.667
(d) 1.0 ha/h

Q125. Width to give TFC 0.667 ha/h at 4 km/h:

(a) 1.2 m (b) 1.67 m

(c) 2.5 m (d) 3.3 m

Answers

Q122 — (b) Rising repair cost + obsolescence exceed value of remaining service

Q123 — (b) 0.5 — *EFC = 40/80 = 0.5 ha/h.*

Q124 — (c) 0.667 — *TFC = 0.5/0.75 = 0.667 ha/h.*

Q125 — (b) 1.67 m — *W = 10×TFC/S = 10×0.667/4 = 1.67 m.*

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.